

Experimental Environment Configuration (Windows)

Linux is a freely available operating system that is highly suitable for operating system development. In the operating system laboratory course, we use Ubuntu 24.04 LTS as the standard experimental environment.

This section will introduce how to set up the experimental environment required for this course on a Windows system.

Configuring an Ubuntu Virtual Machine

In this section, we will guide students to install VMware Workstation Pro on a Windows system and set an Ubuntu virtual machine within VMware.

Installation of VMware Workstation Pro 16

You can download VMware Workstation Pro 17 through the following link and install it.

[Download VMware Workstation Pro](#)

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📁	com.vmware.fusion.zip
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📁	debian-12.9.0-amd64-netinst.iso
📁	OSLab-Ubuntu-24.04-LTS-VMware-ovf.zip
📁	ubuntu-24.04.1-desktop-amd64.iso
⚙️	VMware-workstation-17.6.2-24409262.exe
📁	VMware-workstation-17.6.2-24409262.exe.sha1sum

Installation of Ubuntu24.04

Next, regarding the installation of Ubuntu 24.04, we offer two options. You can choose to download the Ubuntu system image that already comes with the experimental environment, or the official original version of the Ubuntu image.

1. Import the Ubuntu system image with the experimental environment.

This step involves importing a system image with the experimental environment already set up in VMware. If you already have an existing environment, you can refer to the steps in "Manually Configuring the Development Environment" to set up the experimental environment on your system. However, we strongly recommend using the environment we provide to avoid

discrepancies in the experiment caused by system version differences.

First, download the OVF (Open Virtualization Format) file for Ubuntu 24.04 using the following link and then extract it.

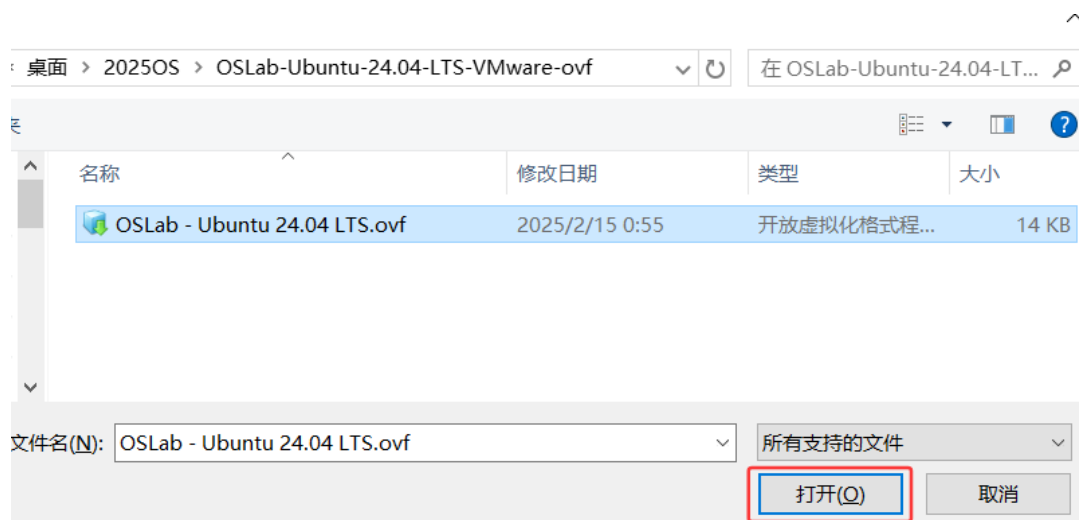
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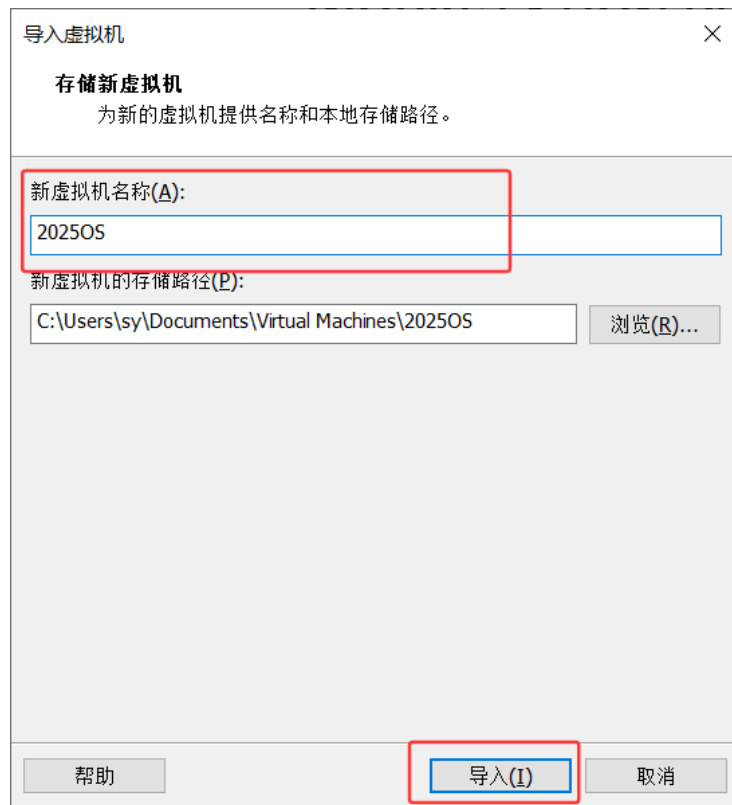
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Name ▾	
←	dl.cra.moe
📁	com.vmware.fusion.zip
📁	com.vmware.fusion.zip.sha1sum
📁	debian-12.9.0-amd64-netinst.iso
📁	OSLab-Ubuntu-24.04-LTS-VMware-ovf.zip
📁	ubuntu-24.04.1-desktop-amd64.iso
⚙️	VMware-workstation-17.6.2-24409262.exe
📁	VMware-workstation-17.6.2-24409262.exe.sha1sum

Next, open VMware and select **File** -> **Open**, locate the OVF file and choose to open it. After entering a name for the virtual machine, wait for the import process to complete.





2. installation of the original ubuntu 24.04

You can download the original Ubuntu 24.04 image from the following link and install it on their own virtual machine.

[Download Ubuntu 24.04](#)

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📁	com.vmware.fusion.zip
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⚙️	VMware-workstation-17.6.2-24409262.exe
📁	VMware-workstation-17.6.2-24409262.exe.sha1sum

After downloading the image, open VMware Workstation and select "**Create a New Virtual Machine**":

WORKSTATION PRO™ 17

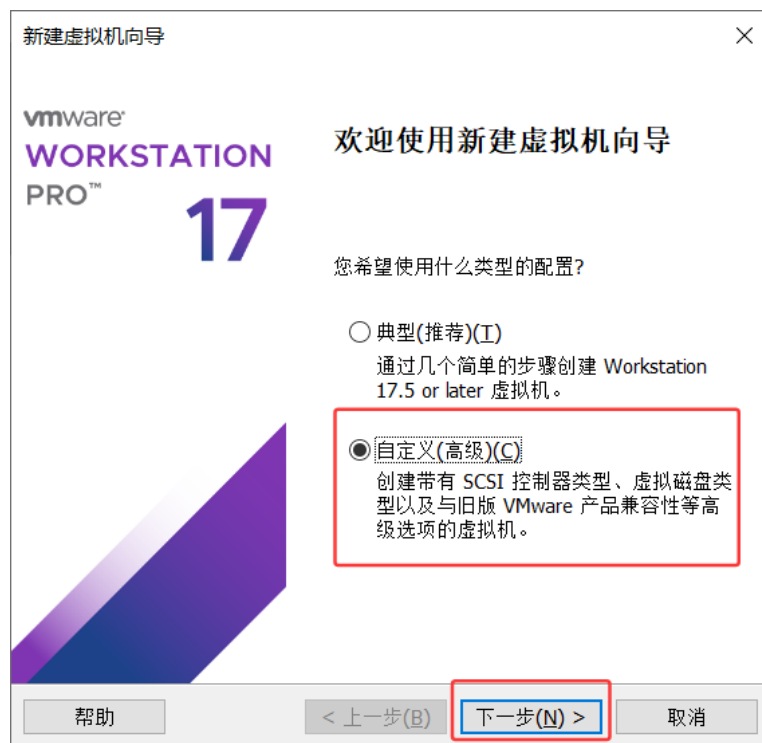


Follow the guidance below to complete the virtual machine configuration:

Note

Students are advised to meet the following requirements during system installation:

1. Language: Try to select English; Chinese paths may lead to issues in completing the experimental content.
2. CPU: Preferably choose 4 cores or more.
3. Virtual machine disk space: It is recommended to allocate 30GB or more for the virtual machine.



新建虚拟机向导

选择虚拟机硬件兼容性

该虚拟机需要何种硬件功能?

虚拟机硬件兼容性

硬件兼容性(H):

Workstation 17.5 or later

兼容:

☒ ESX Server(S)

兼容产品:

Fusion 13.5 or later

Workstation 17.5 or later

限制:

128 GB 内存

32 个处理器

10 个网络适配器

8 TB 磁盘大小

8 GB 共享图形内存

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

安装客户机操作系统

虚拟机如同物理机，需要操作系统。您将如何安装客户机操作系统?

安装来源:

☐ 安装程序光盘(D):

无可驱动器

☒ 安装程序光盘映像文件(iso)(M):

C:\Users\sy\Desktop\2025OS\ubuntu-24.04.1-deskto

浏览(B)...

已检测到 Ubuntu 64 位 24.04.1。

该操作系统将使用简易安装。[\(这是什么?\)](#)

☐ 稍后安装操作系统(S)。

创建的虚拟机将包含一个空白硬盘。

帮助

< 上一步(B)

下一步(N) >

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新建虚拟机向导

✕

简易安装信息

这用于安装 Ubuntu 64 位。

个性化 Linux

全名(F): OS

用户名(U): sy22221111

密码(P):

确认(C):

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

✕

命名虚拟机

您希望该虚拟机使用什么名称?

虚拟机名称(V): Ubuntu 64 位

位置(L): C:\Users\sy\Documents\Virtual Machines\Ubuntu 64 位

浏览(B)...

在“编辑”>“首选项”中可更改默认位置。

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

处理器配置

为此虚拟机指定处理器数量。

处理器

处理器数量(P): 4

每个处理器的内核数量(C): 1

处理器内核总数: 4

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

此虚拟机的内存

您要为此虚拟机使用多少内存?

指定分配给此虚拟机的内存量。内存大小必须为 4 MB 的倍数。

128 GB

64 GB

32 GB

16 GB

8 GB

4 GB

2 GB

1 GB

512 MB

256 MB

128 MB

64 MB

32 MB

16 MB

8 MB

4 MB

此虚拟机的内存(M): 4096 MB

最大推荐内存: 6.2 GB

推荐内存: 2 GB

客户机操作系统最低推荐内存: 1 GB

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

网络类型

要添加哪类网络？

网络连接

☐ 使用桥接网络(R)

为客户机操作系统提供直接访问外部以太网网络的权限。客户机在外部网络上必须有自己的 IP 地址。

☒ 使用网络地址转换(NAT)(E)

为客户机操作系统提供使用主机 IP 地址访问主机拨号连接或外部以太网网络连接的权限。

☐ 使用仅主机模式网络(H)

将客户机操作系统连接到主机上的专用虚拟网络。

☐ 不使用网络连接(I)

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

选择 I/O 控制器类型

您希望将哪种 SCSI 控制器类型用于 SCSI 虚拟磁盘？

I/O 控制器类型

SCSI 控制器:

☐ BusLogic(U)

(不适用于 64 位客户机)

☒ LSI Logic(L)

(推荐)

☐ LSI Logic SAS(S)

☐ 准虚拟化 SCSI(P)

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

选择磁盘类型

您要创建何种磁盘？

虚拟磁盘类型

☐ IDE(I)

☒ SCSI(S) (推荐)

☐ SATA(A)

☐ NVMe(V)

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

选择磁盘

您要使用哪个磁盘？

磁盘

☒ 创建新虚拟磁盘(V)

虚拟磁盘由主机文件系统上的一个或多个文件组成，客户机操作系统会将其视为单个硬盘。虚拟磁盘可在一台主机上或多台主机之间轻松复制或移动。

☐ 使用现有虚拟磁盘(E)

选择此选项可重新使用以前配置的磁盘。

☐ 使用物理磁盘 (适用于高级用户)(P)

选择此选项可为虚拟机提供直接访问本地硬盘的权限。需要具有管理员特权。

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

指定磁盘容量

磁盘大小为多少?

最大磁盘大小 (GB)(S):

30.0

针对 Ubuntu 64 位 的建议大小: 20 GB

☐ 立即分配所有磁盘空间(A)。

分配所有容量可以提高性能，但要求所有物理磁盘空间立即可用。如果不立即分配所有空间，虚拟磁盘的空间最初很小，会随着您向其中添加数据而不断变大。

☐ 将虚拟磁盘存储为单个文件(Q)

☒ 将虚拟磁盘拆分成多个文件(M)

拆分磁盘后，可以更轻松地在计算机之间移动虚拟机，但可能会降低大容量磁盘的性能。

帮助

< 上一步(B)

下一步(N) >

取消

新建虚拟机向导

指定磁盘文件

您要在何处存储磁盘文件?

磁盘文件(E)

将使用多个磁盘文件创建一个 30 GB 虚拟磁盘。将根据此文件名自动命名这些磁盘文件。

Ubuntu 64 位.vmdk

浏览(R)...

帮助

< 上一步(B)

下一步(N) >

取消



After that, run the virtual machine to complete the installation.

Note

Since the original Ubuntu system does not include the development environment for the experiments in this course, you will also need to refer to the section "Manual Configuration of the Development Environment" to set up the experimental development environment. Additionally, we recommend installing VSCode for code reading and editing. These configurations are already included in the OVF file..

Manual Configuration of the Development Environment

If you use the OVF image that we have provided, **you can skip this part.**

By this step, please make sure that you have already set up the Linux environment and are able to open a Terminal to execute commands.

Warning

Please use the latest available release version whenever possible. The following code is based on Ubuntu 24.04 LTS.

Install the GCC toolchain and QEMU

Use the package manager `apt` to install dependencies. Execute in the terminal:

```
sudo apt update && sudo apt install gcc-riscv64-unknown-elf qemu-system-misc git  
make cmake python3-pip elfutils gdb-multiarch
```

After installation, run the following commands to check the versions: `riscv64-unknown-elf-gcc --version`, `qemu-system-riscv64 --version`, and `gdb-multiarch --version`.

```
Platform Name      : riscv-virtio,qemu
Platform Features  : medeleg
```

```

Platform HART Count      : 1
Platform IPI Device      : aclint-mswi
Platform Timer Device    : aclint-mtimer @ 100000000Hz
Platform Console Device  : uart8250
Platform HSM Device      : ---
Platform PMU Device      : ---
Platform Reboot Device   : syscon-reboot
Platform Shutdown Device : syscon-poweroff
Platform Suspend Device  : ---
Platform CPPC Device     : ---
Firmware Base            : 0x80000000
Firmware Size            : 327 KB
Firmware RW Offset       : 0x40000
Firmware RW Size         : 71 KB
Firmware Heap Offset     : 0x49000
Firmware Heap Size       : 35 KB (total), 2 KB (reserved), 11 KB (used), 21 KB (free)
Firmware Scratch Size    : 4096 B (total), 416 B (used), 3680 B (free)
Runtime SBI Version      : 2.0

Domain0 Name             : root
Domain0 Boot HART        : 0
Domain0 HARTs            : 0*
Domain0 Region00         : 0x000000000100000-0x000000000100fff M: (I,R,W)
S/U: (R,W)
Domain0 Region01         : 0x000000001000000-0x000000001000fff M: (I,R,W)
S/U: (R,W)
Domain0 Region02         : 0x000000000200000-0x000000000200fff M: (I,R,W)
S/U: ()
Domain0 Region03         : 0x000000008004000-0x000000008005fff M: (R,W) S/U:
()
Domain0 Region04         : 0x000000008000000-0x000000008003fff M: (R,X) S/U:
()
Domain0 Region05         : 0x000000000c40000-0x000000000c5ffff M: (I,R,W)
S/U: (R,W)
Domain0 Region06         : 0x000000000c00000-0x000000000c3ffff M: (I,R,W)
S/U: (R,W)
Domain0 Region07         : 0x000000000000000-0xffffffff M: () S/U:
(R,W,X)
Domain0 Next Address     : 0x000000008020000
Domain0 Next Arg1        : 0x000000009fe0000
Domain0 Next Mode        : S-mode
Domain0 SysReset         : yes
Domain0 SysSuspend       : yes

Boot HART ID             : 0
Boot HART Domain         : root
Boot HART Priv Version   : v1.12
Boot HART Base ISA       : rv64imafdc
Boot HART ISA Extensions : sstc,zicntr,zihpm,zicboz,zicbom,sdtrig,svadu
Boot HART PMP Count      : 16
Boot HART PMP Granularity : 2 bits
Boot HART PMP Address Bits : 54
Boot HART MHPM Info      : 16 (0x0007fff8)
Boot HART Debug Triggers : 2 triggers
Boot HART MIDELEG        : 0x000000000001666
Boot HART MEDELEG        : 0x000000000f0b509

```

=====

Hello world!

=====

Boot stack: 0x000000008021d000

clean bss: 0x000000008021e000 - 0x0000000080228000

Boot m_hartid 0

[INFO 0,-1] bootcpu_entry: basic smp init, thread_id available now, we are
cpu 0: 0x00000000802270d8

Kernel Starts Relocating...

Kernel size: 0x0000000000028000, Rounded to 2MiB: 0x0000000000200000

[INFO 0,-1] bootcpu_start_relocation: Kernel phy_base: 0x0000000080200000,
phy_end_4k:0x0000000080228000, phy_end_2M 0x0000000080400000

Mapping Identity: 0x0000000080200000 to 0x0000000080200000

Mapping kernel image: 0xffffffff80200000 to 0x0000000080200000

Mapping Direct Mapping: 0xffffffffc080400000 to 0x0000000080400000

Enable SATP on temporary pagetable.

Boot HART Relocated. We are at high address now! PC: 0xffffffff80203cc4

[INFO 0,-1] kvm_init: boot-stage page allocator: base 0xffffffffc080400000, end
0xffffffffc080600000

[INFO 0,-1] kvmmake: Memory after kernel image (phys) size = 0x0000000003c00000

[INFO 0,-1] kvm_init: enable pageing at 0x8000000000080400

[INFO 0,-1] kvm_init: boot-stage page allocator ends up: base

0xffffffffc080400000, used: 0xffffffffc080411000

Relocated. Boot halt sp at 0xffffffffffff001fb0

Boot another cpus.

- booting hart 1: hsm_hart_start(hartid=1, pc=_entry_sec, opaque=1) = -3.

waiting for hart online

skipped for hart 1

- booting hart 2: hsm_hart_start(hartid=2, pc=_entry_sec, opaque=1) = -3.

waiting for hart online

skipped for hart 2

- booting hart 3: hsm_hart_start(hartid=3, pc=_entry_sec, opaque=1) = -3.

waiting for hart online

skipped for hart 3

System has 1 cpus online

UART init.

[INFO 0,-1] kpgmgrinit: page allocator init: base: 0xffffffffc080411000, stop:
0xffffffffc084000000

[INFO 0,-1] allocator_init: allocator mm init base 0xffffffffd00000000

[INFO 0,-1] allocator_init: allocator vma init base 0xffffffffd01000000

[INFO 0,-1] allocator_init: allocator proc init base 0xffffffffd02000000

[INFO 0,-1] allocator_init: allocator kstrbuf init base 0xffffffffd03000000

applist:

init

sh

test_arg

test_malloc

[INFO 0,-1] load_init_app: load init proc init

[INFO 0,-1] bootcpu_init: start scheduler!

init: starting sh